Bayesian Time-Varying Multi-Seasonal ARMA Models

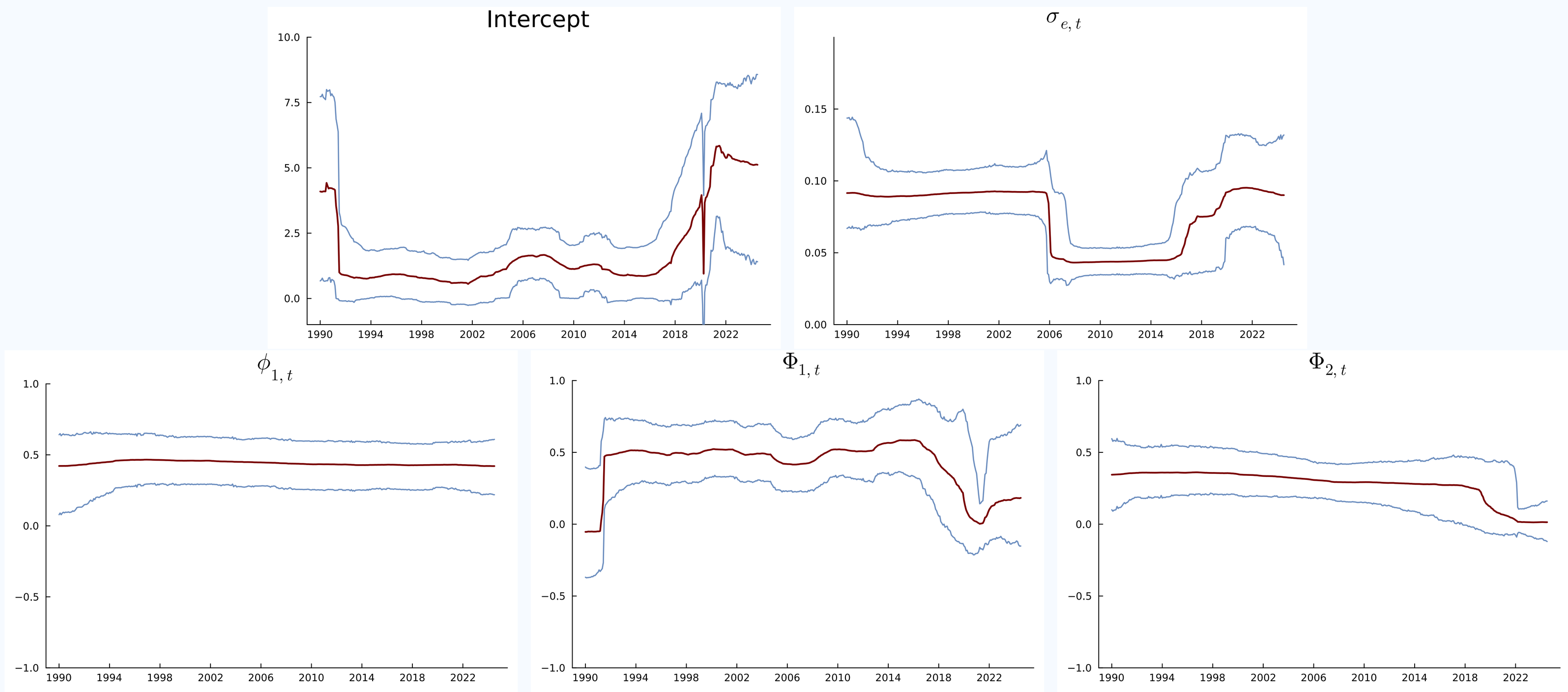
Stable and Scalable Inference for Nonstationary Time Series

Ganna Fagerberg, Mattias Villani, Robert Kohn | ISBA 2026

Stockholm University

1. Why TV-SARMA Models?

Static Model (Box & Jenkins) $\Longrightarrow$ **TV Model (Box & Jenkins 2.0)**



US Air Passenger Data: Posterior TV-SAR$(1,2)_{12}$ medians and 95% credible bands.

TV-SAR captures how major events, such as the COVID-19 pandemic, alter trend, seasonality and volatility over time. In contrast, static models assume fixed dynamics.

2. Modeling Framework and Contributions

Bayesian | **TV-SARMA** | **Multiple Seasons** | **Stable & Invertible**
DSP Priors | **Nonlinear Inference** | **Adaptive Filtering** | **Scalable**

3. State-Space Representation

Observation: $y_t = \boldsymbol{x}_{0t}^\top \tilde{\phi}_t^{(AR)} + \boldsymbol{z}_{0t}^\top \tilde{\psi}_t^{(MA)} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma^2)$

Constraint: $\tilde{\phi}_t, \tilde{\psi}_t = \tilde{\boldsymbol{g}}(\theta_t)$ **(stability & invertibility)**

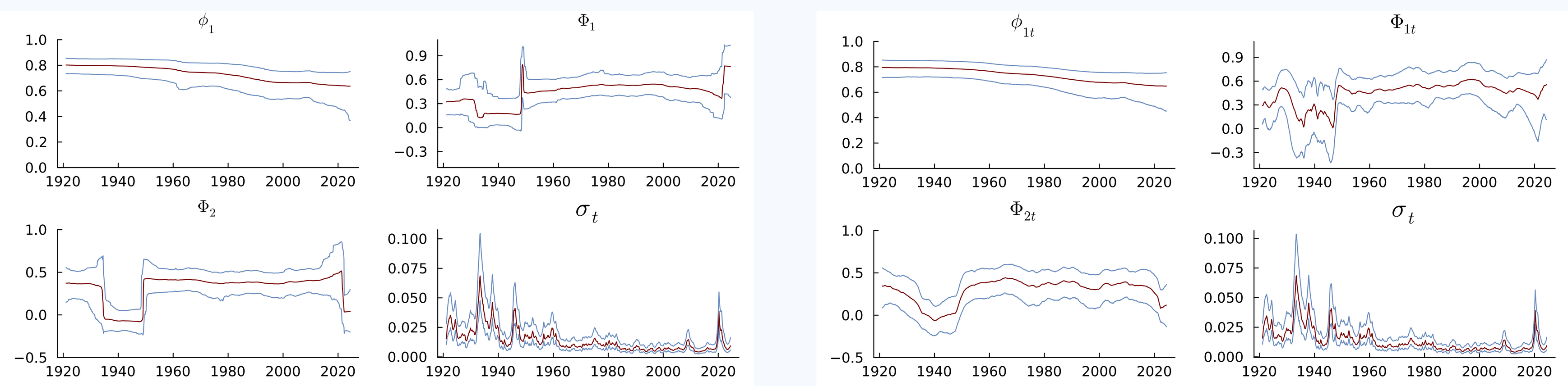
State: $\theta_t = \theta_{t-1} + \nu_t, \qquad \nu_t \sim N(0, \mathrm{Diag}\{\exp(\boldsymbol{h}_t)\})$

DSP: $\boldsymbol{h}_t = \mu + \kappa(\boldsymbol{h}_{t-1} - \mu) + \eta_t, \qquad \eta_t \sim Z(1/2, 1/2, 0, 1).$

The observation equation is the multiplicative SARMA model in nonlinear regression form. Here, $\boldsymbol{x}_{0t}$ and $\boldsymbol{z}_{0t}$ collect the nonzero lagged observations and innovations (including presample and initial values), while $\tilde{\phi}_t$ and $\tilde{\psi}_t$ contain the corresponding nonzero AR and MA coefficients, including interaction terms.

4. Dynamic Shrinkage vs. Gaussian Regularization

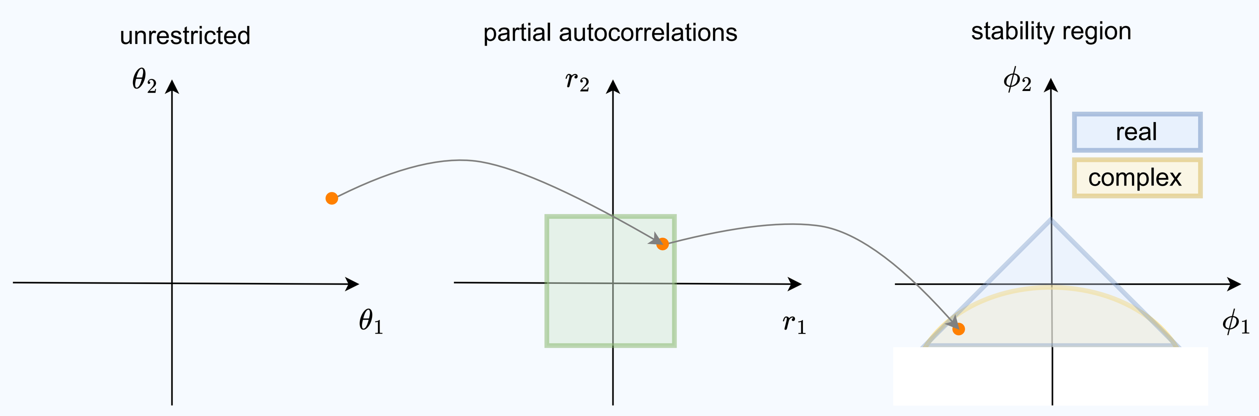
DSP Prior with SV | **Gaussian Prior with SV**



US Industry Production Index Data: Posterior TV-SAR$(1,2)_{12}$ medians and 95% credible bands.

DSP prior (left): adaptive shrinkage, smoother trajectories, and abrupt-change detection.
Gaussian prior (right): less regularization, more variable trajectories.

5. Stability Parametrization and Uniform Prior



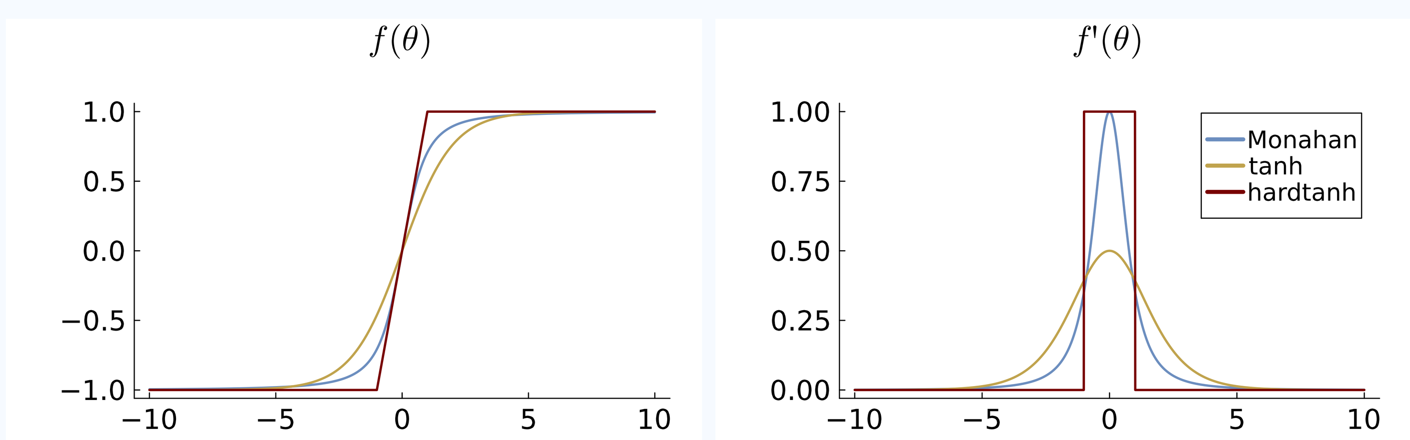
$$\theta \xrightarrow{\text{Monahan}} r \xrightarrow{\text{Levinson–Durbin}} \phi$$

Uniform prior over the stable AR region with Monahan (1984)

Same idea applies to ensure invertibility!

6. Robust Stability Parametrization

Monahan (1984) $\Longrightarrow$ tanh $\Longrightarrow$ **hardtanh**

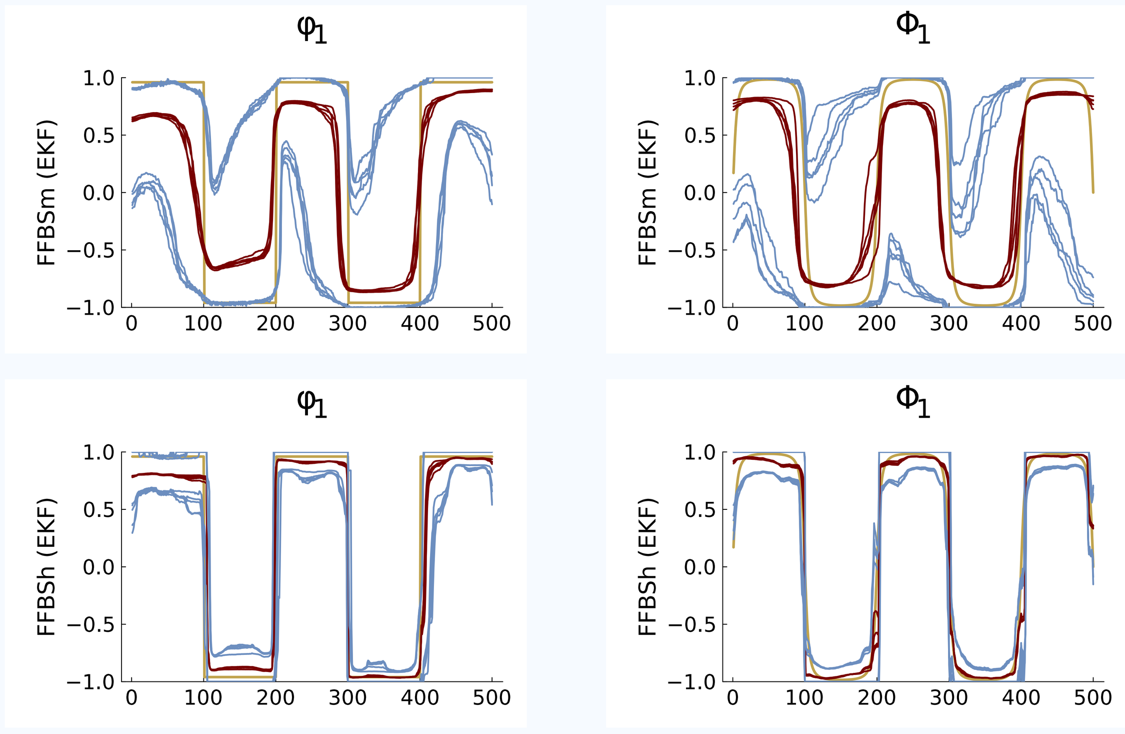


$$\frac{\theta}{\sqrt{1+\theta^2}} \longrightarrow \tanh(\theta/2) \longrightarrow \min(1, \max(-1, \theta))$$

(*vanishing gradients*) (*vanishing gradients*) (***linear interior***)

7. Near the Stability Boundary

TV-SAR$(1,1)_{12}$ | **Extreme Boundary Dynamics** | **Simulated Data**



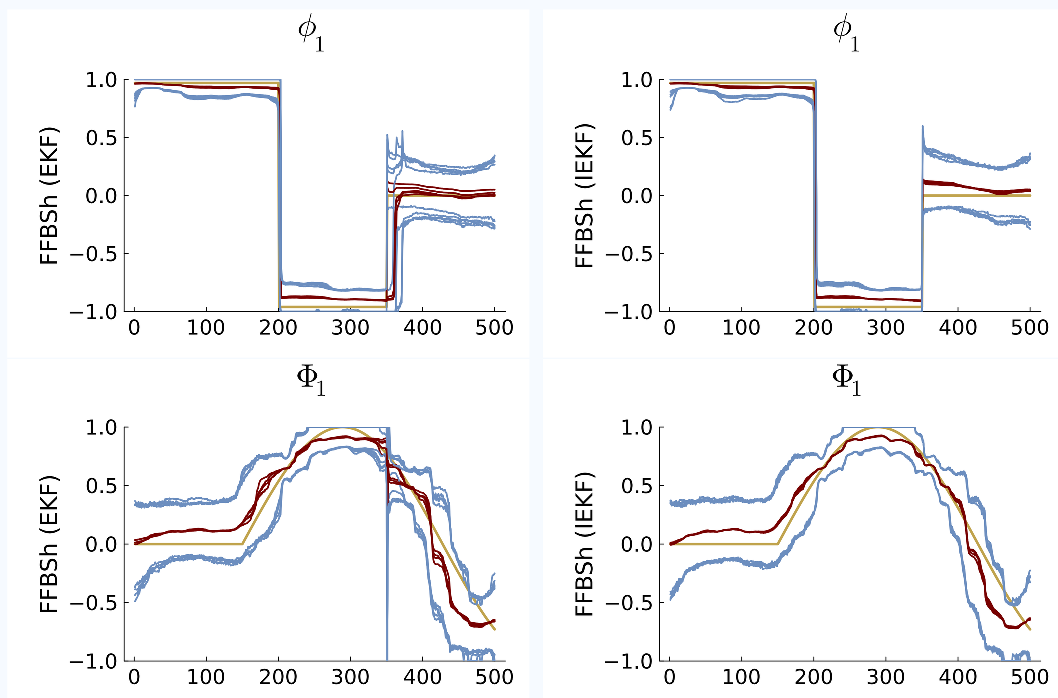
Posterior medians and 95% credible bands from five independent runs under **Monahan** (first row) and **hardtanh** (second row) parametrizations.

Monahan (first row): unstable filtering and inflated credible bands.
Hardtanh + constrained filtering (second row): robust, numerically stable inference.

8. Robust Adaptive Inference

Hardtanh | **Constrained Filtering** | **Adaptive Inference**

$$\textbf{FFBS+EKF} \ (\textit{Fast}) \xrightarrow{\text{NIS trigger}} \textbf{FFBS+IEKF} \ (\textit{Accurate})$$



Simulated TV-SAR$(1,1)_{12}$: posterior medians and 95% credible bands from five independent runs under **FFBS+EKF** (left column) and **FFBS+IEKF** (right column).

Adaptive inference: fast when possible, accurate when necessary!

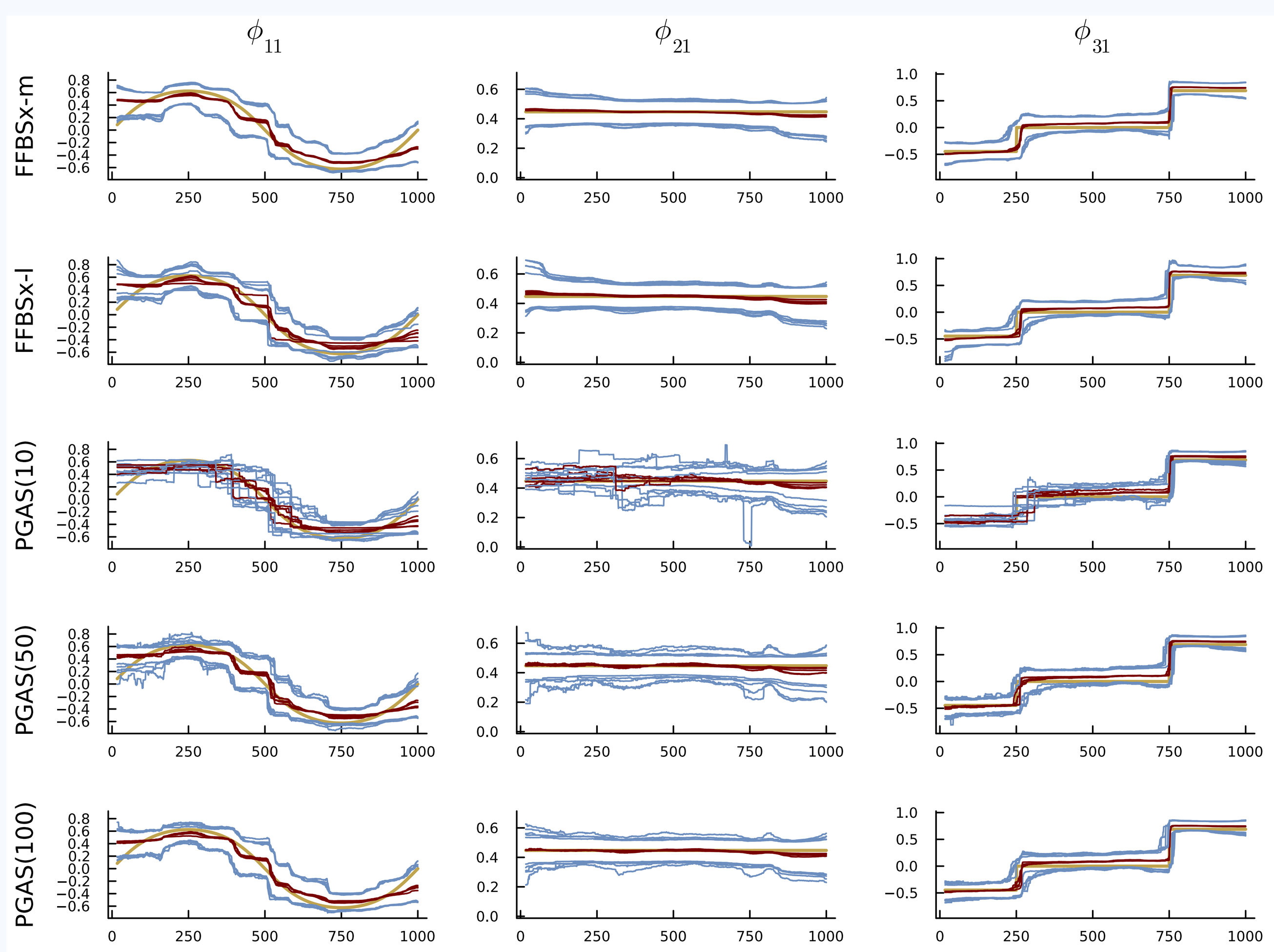
9. Gibbs sampling scheme

States (FFBS+(I)EKF) $\Longrightarrow$ **DSP** (Kowal, 2017) $\Longrightarrow$ **Presample** (Closed-form) $\Longrightarrow$ **Errors** (Closed-form) $\Longrightarrow$ **Volatility** (SV+DSP)

10. Scaling to Multiple Seasonalities

Grouping (*Fewer state updates*) $\Longrightarrow$ **Sequential EKF** (*Lower filtering cost*) $\Longrightarrow$ **Scalable Inference** (*Long high-frequency series*)

11. Simulations



TV-SAR(1,1,1)$_{(4,12)}$: Posterior medians (red) and 95% credible bands (blue) from five independent runs.

Scan for article and poster

Contact: ganna.fagerberg@stat.su.se Article: https://www.tandfonline.com/doi/full/10.1080/10618600.2026.2614102